DECK 17 · WEEK 7 · MORNING 1

Derivatives I
Forwards, Futures & Swaps

Firm-commitment instruments and their no-arbitrage pricing

CFA Level I · Vol 7 Learning Modules 1–7

Roadmap — Derivatives I: Forwards, Futures & Swaps

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Derivative markets & instruments

Definition · OTC vs ETD · ISDA · central clearing · firm commitments vs contingent claims

02

Benefits, risks & uses

Risk transfer · price discovery · ops efficiency · leverage / basis / liquidity / credit risk · issuer vs investor

03

Arbitrage, replication & cost of carry

Law of one price · cash-and-carry arbitrage · replication identity · dividends / storage / income adjustments

04

Forwards & futures

$F_0(T)=S_0(1+r)^T$ · MTM over life · equity / FX / index forwards · IRP · daily margining · basis · convexity bias

05

Swaps

Strip of forwards · par swap rate · PV(floating)=PV(fixed) · IRS = long FRN + short fixed bond · valuation over time

Learning objectives

By the end of this deck you should be able to:

1. Describe the basic features of a derivative and contrast OTC vs exchange-traded markets, including the role of ISDA, central clearing and margining.

2. Distinguish firm commitments (forwards, futures, swaps) from contingent claims (options, credit derivatives) by their payoff profile and settlement mechanics.

3. Evaluate the benefits (risk transfer, price discovery, operational efficiency) and risks (leverage, basis, liquidity, counterparty credit) of derivative use.

4. Apply the law of one price to price a forward via cash-and-carry arbitrage and adjust the formula for income, storage costs, and interest-rate differentials.

5. Price and value forward contracts on equities (with discrete dividends and dividend yields), FX (via covered interest-rate parity), and interest rates (FRAs).

6. Price and value plain-vanilla interest-rate swaps by solving for the par swap rate and by interpreting the swap as a pair of offsetting bond positions.

SECTION 01

Derivative markets & instruments

A derivative is a contract whose value is derived from an underlying — the buyer gets a long-like exposure, the seller a short-like exposure, at a specified maturity

A forward on 1,000 Airbus shares at EUR 30/share settles either in physical delivery or in cash for the spot-vs-forward difference.

$$\text{Payoff at } T \text{ (long)} = S_T - F_0(T) \qquad \text{Payoff at } T \text{ (short)} = F_0(T) - S_T$$

where: S_T = underlying spot price at maturity; $F_0(T)$ = forward price agreed at inception ($t = 0$).

UNDERLYING

Equities, fixed income (bonds, interest rates), FX, commodities, credit, volatility. May be a basket or an index, not necessarily a single asset.

COUNTERPARTIES

Legal entities with matching obligations. Almost always one end-user (issuer or investor) and a financial intermediary. Counterparty credit risk is a first-order concern.

CONTRACT SIZE & MATURITY

Notional (or contract size) agreed at outset. Maturity = time to settlement. Cash settlement = net difference; physical settlement = exchange of asset for cash.

Source: CFA Institute, Curriculum Vol 7, Learning Module 1, "Derivative Features", pp. 5–8.

OTC derivatives are customized bilateral contracts under an ISDA; exchange-traded derivatives are standardized, centrally cleared, and margined daily

Dodd-Frank and EMIR pushed most OTC rate and FX swaps into central clearing, narrowing the structural gap with ETDs.

Feature	Over-the-counter (OTC)	Exchange-traded (ETD)
Contract terms	Customized (size, maturity, underlying, delivery)	Standardized contract spec
Counterparty credit risk	Bilateral — mitigated by ISDA + CSA collateral	Borne by the CCP (novation)
Settlement	Typically at maturity	Daily mark-to-market via margin
Transparency / liquidity	Lower; dealer quotes, negotiated	Higher; public order book
Typical products	FX forwards, IRS, credit default swaps	Futures, listed options
Regulation	Post-2010 central clearing for standardized OTC	Exchange + CFTC / ESMA / equivalent

ISDA MASTER AGREEMENT

A standardized legal shell covering netting, default, early termination and collateral (via the Credit Support Annex). Enables one agreement to cover thousands of trades between two counterparties — the plumbing of OTC derivative markets.

Source: CFA Institute, Curriculum Vol 7, Learning Module 1, "Derivative Markets", pp. 14–18.

Firm commitments fix the terms today and settle in the future; contingent claims give the buyer an optional payoff — forwards, futures and swaps vs options

Firm commitments have SYMMETRIC payoffs (buyer's gain = seller's loss). Contingent claims have ASYMMETRIC payoffs (buyer caps loss at premium).

FIRM COMMITMENTS

Both parties are OBLIGED to transact at the pre-agreed terms.

- Forward — OTC, one-shot, settled at maturity
- Future — ETD, one-shot at maturity but daily MTM in cash
- Swap — OTC, periodic exchange (usually IRS fixed vs MRR)

$V_0(T) = 0$ at inception (no cash changes hands).
Payoff is SYMMETRIC: $V_T(\text{long}) = S_T - F_0(T)$.

CONTINGENT CLAIMS

ONE party (the buyer) decides whether and when to exercise.

- Call option — right to buy at X
- Put option — right to sell at X
- Credit derivative (CDS) — payoff contingent on a credit event

Buyer pays a PREMIUM (c_0 or p_0) at inception → $V_0 \neq 0$.
Payoff is ASYMMETRIC: buyer's loss is capped at premium.

Source: CFA Institute, Curriculum Vol 7, Learning Module 2, "Forward Commitment vs Contingent Claim", pp. 26–49.

Forwards, futures and swaps share a symmetric payoff profile and zero value at inception — they differ in settlement mechanics and in the cash-flow shape

Same underlying economic exposure to the forward curve — very different daily cash-flow and credit-risk profiles.

Feature	Forward	Futures	Swap
Market	OTC	Exchange-traded	OTC (centrally cleared std)
Contract	Customized	Standardized	Customized
Cash flow profile	Single exchange at T	Single economic exchange + daily MTM	Periodic exchange over life
Credit risk	Counterparty (bilateral)	CCP (very low)	Counterparty + margin
Initial value	$V_0 = 0$	$V_0 = 0$ (resets daily)	$V_0 = 0$
Payoff shape	Symmetric	Symmetric	Symmetric (strip of forwards)

MENTAL MODEL

A swap is economically a STRIP of forwards, but priced at a SINGLE fixed rate (the par swap rate) rather than at each period's own forward rate. A futures contract is a forward wrapped in a daily settlement mechanism that eliminates most counterparty credit risk.

Source: CFA Institute, Curriculum Vol 7, Learning Module 2, "Forwards, Futures, and Swaps", pp. 26–37.

SECTION 02

Benefits, risks & uses

Derivatives deliver four core benefits — risk transfer, price discovery, operational efficiency, and market efficiency — that cash markets alone cannot match

Same underlying exposure as the cash market, a fraction of the capital, and a bridge between economic decisions and the moment they can be transacted.

Benefit	Mechanism	CFA vignette
Risk allocation / transfer	Bridge the timing gap between an economic decision and the ability to transact	Montau FX hedge
Price discovery	Derivative prices reveal expectations: index futures, rate futures, implied vol	Index futures open
Operational efficiency	Lower transaction costs, higher liquidity, small upfront cash, easy shorting	Procam gold margin
Market efficiency	Cheaper to exploit mispricings → prices converge on fundamentals faster	Arbitrage enforcement

OPERATIONAL EFFICIENCY — PROCAM GOLD FUTURES VS SPOT

Same 100-oz long-gold exposure for 3 months; 5% funding cost.

Futures: USD 4,950 margin → 3m interest USD 61.88. Spot: USD 177,000 → 3m interest USD 2,212.50. 36× capital efficiency.

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Derivative Benefits", pp. 55–62.

The same leverage that makes derivatives efficient also creates hidden risks — leverage, basis, liquidity, and counterparty credit risk demand explicit management

The tools are powerful because they are leveraged; the same property makes unmanaged positions a route to systemic stress.

1. **Implicit leverage**
   Small margin → large notional. A 1% move can wipe out the margin — derivatives are often the cheapest way to take risk and the fastest way to blow up.

2. **Basis risk**
   The derivative's underlying does not perfectly match the hedged exposure (e.g. WTI future hedging Brent cargo). Residual P&L = basis × notional.

3. **Liquidity risk**
   Mismatch in cash-flow TIMING between hedge and hedged item: daily margin calls on a 5-year futures hedge vs annual coupon — can force early unwind.

4. **Counterparty credit risk**
   Risk the dealer fails to honor an OTC contract. Mitigated by CCP clearing, ISDA master + CSA collateral, netting.

[1] Excessive derivative leverage was a catalyst in the 2008 financial crisis (AIG CDS, Lehman bilateral exposures) — motivating post-crisis central clearing mandates.

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Derivative Risks", pp. 63–66.

Issuers use derivatives to hedge commercial and financing exposures; investors use them to modify portfolio cash flows and replicate exposures

Issuers typically seek HEDGE ACCOUNTING (fair-value, cash-flow, or net-investment hedge) to minimize P&L volatility from the derivative itself.

ISSUER USE CASES

- FX export hedge (Montau KRW/EUR forward to lock revenue in EUR)
- Commodity input hedge (airline buying jet-fuel swaps)
- Convert floating-rate debt to fixed via pay-fixed IRS → cash-flow hedge accounting
- Convert fixed-rate debt to floating via receive-fixed IRS → fair-value hedge accounting
- Pre-issuance rate lock via forward-starting swap

INVESTOR USE CASES

- Modify portfolio duration via receive-fixed IRS (longer) or pay-fixed IRS (shorter) — no bond trading
- Covered call on existing long equity → extra yield, capped upside
- Synthetic index exposure via futures (low capital, high liquidity)
- Hedge FX exposure on foreign equities / bonds
- Express credit view via CDS without holding the bond

Source: CFA Institute, Curriculum Vol 7, Learning Module 3, "Issuer/Investor Use of Derivatives", pp. 67–72.

SECTION 03

Arbitrage, replication & cost of carry

Derivative pricing rests on a single engine — the law of one price enforced by riskless arbitrage between spot and forward positions

If two positions deliver the same future cash flows, they must have the same price today. Any deviation is arbitraged away in seconds.

$$\text{Law of one price:} \quad S_0 = F_0(T) / (1 + r)^T \quad \Leftrightarrow \quad F_0(T) = S_0 (1 + r)^T$$

where: S_0 = spot price today; $F_0(T)$ = forward price agreed today for delivery at T; r = risk-free rate over horizon T

ARBITRAGE

A trade that makes a RISKLESS positive profit with ZERO net investment. Requires simultaneous positions. In efficient markets, arbitrage is fleeting — but enforces the no-arb condition.

REPLICATION

Reproduce a derivative's cash flows with a spot position + risk-free borrowing/lending. Long forward ≡ buy spot + borrow PV(F). If replication and derivative prices differ, arbitrage opens.

COST OF CARRY

Net cost of holding the underlying from 0 to T: opportunity cost r, plus C (storage), minus I (income earned). The forward price must cover this carry to prevent arbitrage.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, "Arbitrage and Replication", pp. 80–87.

If a forward is mispriced relative to the no-arbitrage spot-plus-carry, a cash-and-carry trade locks in a riskless profit — the worked example that pins $F_0(T)$

Sell the rich leg, buy the cheap leg, carry to maturity — the textbook arbitrage mechanic that every pricing formula in this section enforces.

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No-arb:  F_0(T) = S_0 (1+r)^T     Observed F > No-arb ⇒ SELL
              forward  +  BUY spot  +  BORROW S_0
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CFA LM 4 PP#4 — CASH-AND-CARRY ARBITRAGE

S_0 = USD 100; observed 3-month forward F_0 = USD 102; risk-free rate r = 5% p.a.; T = 0.25y. Underlying has no dividend or storage.

1. Compute the no-arbitrage forward: F*_0(T) = 100 × (1.05)^0.25 = USD 101.23.
2. Observed 102 > 101.23 ⇒forward is RICH. Sell forward at 102, buy spot at 100, finance purchase by borrowing 100 at 5%.
3. At t = T: deliver the spot asset (held) and receive USD 102 under the forward.
4. Repay the loan: 100 × (1.05)^0.25 = USD 101.23. Keep the difference.
5. Arbitrage profit = 102 − 101.23 = USD 0.77 per contract, riskless, zero net investment at t = 0.

Riskless profit USD 0.77 pins the forward at 101.23 — this arbitrage is the proof of $F_0(T) = S_0(1+r)^T$.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Practice Problem 4, pp. 87.

A long forward commitment can be replicated by borrowing at r to buy the underlying at spot — the replication identity that gives $F_0(T) = S_0(1+r)^T$

Long forward ≡ LONG asset (financed) ; short forward ≡ SHORT asset (lend proceeds). This identity is why a forward has $V_0 = 0$.

Position	t = 0 cash flows	t = T cash flows	Net at T
LONG forward	0 (V_0 = 0)	Receive asset, pay $F_0(T)$	$S_T - F_0(T)$
Replicate long: buy spot + borrow	Buy S at S_0; borrow S_0 at r	Own asset worth S_T; repay $S_0(1+r)^T$	$S_T - S_0(1+r)^T$
SHORT forward	0 (V_0 = 0)	Deliver asset, receive $F_0(T)$	$F_0(T) - S_T$
Replicate short: short spot + lend	Short-sell S at S_0; lend S_0 at r	Cover short at S_T; receive $S_0(1+r)^T$	$S_0(1+r)^T - S_T$

THE IDENTITY THAT PRICES EVERY FORWARD

For the two long columns to be indifferent, $S_T - F_0(T)$ must equal $S_T - S_0(1+r)^T$, which forces $F_0(T) = S_0(1+r)^T$. Negative r flips the sign: $F_0 < S_0$. The replication identity extends to any asset once cost of carry is added.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, "Replication", pp. 83–87.

Cost of carry adjusts the replication formula for income earned (I) and costs incurred (C) while holding the underlying — dividends lower F, storage costs raise F

Dividends, coupons, and convenience yields are BENEFITS the spot holder gets that the forward holder does not; storage, insurance, spoilage are added COSTS.

$$\text{Discrete:}\quad F_0(T) = [S_0 - PV(I) + PV(C)]\,(1 + r)^T$$

$$\text{Continuous:}\quad F_0(T) = S_0 \cdot e^{(r + c - i)\,T}$$

where: I, C = income and cost as present values; i, c = income and cost expressed as continuous rates of return; r = risk-free rate

Asset class	Income benefit	Cost	Forward vs spot
Non-dividend equity / zero	None	None	F > S (carry = r)
Dividend-paying stock / index	Dividend I or yield i	None	F depends on (r – i)
Coupon bond	Coupon I	None	F usually > S, less coupons
FX (f/d pair)	Foreign r-free rate earned	Domestic r-free rate paid	Sign of (rf – rd) decides
Physical commodity	Convenience yield (if scarce)	Storage, insurance, spoilage	Often F > S (contango); F < S backwardation

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, "Costs and Benefits of Ownership", pp. 88–97.

SECTION 04

Forwards & futures

At inception the forward price equates the discounted future value of spot plus carry — $V_0(T) = 0$ to both counterparties

No cash changes hands at t=0 because the contract price already embeds the carry. Valuation only becomes non-zero once S, r, or time move.

$$F_0(T) = S_0 (1 + r)^T \qquad V_0(T) = 0 \quad \text{(to both long and short)}$$

where: $F_0(T)$ = pre-agreed forward price; $V_0(T)$ = MTM value at inception; r, T consistent units.

CFA LM 5 EXAMPLE — VFO BIOMIAN FORWARD AT INCEPTION

VFO sells 1,000 Biomian shares forward to an intermediary. Spot S_0 = INR 295; T = 0.5 year; no dividend; risk-free rate r = 4%.

1. Compute forward: $F_0(T) = 295 \times (1.04)^{0.5}$
2. $(1.04)^{0.5} = 1.01980$
3. $F_0(T) = 295 \times 1.01980$ = INR 300.84 per share
4. $V_0(T) = 0$: VFO has not yet gained or lost — the 4% r is already baked in

$F_0(T)$ = INR 300.84 — contract size = 1,000 × 300.84 = INR 300,840.

Source: CFA Institute, Curriculum Vol 7, Learning Module 5, "Pricing vs Valuation", pp. 102–106.

After inception, the MTM of a long forward is the current spot minus the PV of the original forward price — the position re-values as S and r move

Intuition: a forward is a PROMISE to pay F_0 at T; that promise re-values when the PV of F_0 moves away from the current spot.

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V_t(T) long  =  S_t  -  F_0(T) (1 + r)^-(T - t)        V_t(T)
          short  =  F_0(T) (1 + r)^-(T - t)  -  S_t
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where: S_t = spot price now; T − t = remaining life; r = current risk-free rate.

CFA LM 5 EXAMPLE 3 — VFO MTM AT t = 0.25

Biomian short forward (seller): F_0 = INR 300.84, r = 4%, T = 0.5y. At t = 0.25, spot has fallen to S_t = INR 285. Compute VFO's MTM.

1. PV of F_0 at t: 300.84 × (1.04)^−0.25 = 300.84 × 0.99024 = INR 297.90
2. VFO short MTM: V_t = F_0(1+r)^−(T−t) − S_t = 297.90 − 285
3. V_t = INR 12.90 per share → total 1,000 × 12.90 = INR 12,900 gain
4. If r had risen to 8%: PV(F_0) = 300.84 × (1.08)^−0.25 = 295.11 → MTM only 10.11 (higher r shrinks short's gain)

Short MTM gain = INR 12.90/share at r = 4%; falls to INR 10.11/share at r = 8%.

Source: CFA Institute, Curriculum Vol 7, Learning Module 5, "Pricing and Valuation over Life", pp. 106–110.

Equity forwards with discrete dividends price at $F_0(T) = [S_0 - PV(I)](1+r)^T$ — the dividend accrues to the spot holder, not the forward buyer

Rule: subtract the PV of every dividend paid BEFORE maturity from S_0 before capitalizing at r. Dividends paid AFTER T are irrelevant.

$$F_0(T) = [\, S_0 - PV_0(I) \,] \cdot (1 + r)^T$$

where: $PV_0(I)$ = present value at t = 0 of every income (dividend/coupon) received over [0, T].

HIGHTEST / UNILEVER EQUITY FORWARD — T = 0.5y

Hightest delivers 1,000 UL shares in 6 months. S_0 = EUR 50. UL pays quarterly dividend 0.30 at t = 0.25 and t = 0.5. r = 5%.

1. PV(I) = 0.30 × (1.05)^−0.25 + 0.30 × (1.05)^−0.5
2. = 0.2964 + 0.2928 = EUR 0.5892
3. $F_0(T)$ = (50 − 0.5892) × (1.05)^0.5
4. = 49.4108 × 1.02470 = EUR 50.6310 per share
5. Contract value = 1,000 × 50.6310 = EUR 50,631.00

F_0 = EUR 50.6310. Without dividends F_0 would be 50 × (1.05)^0.5 = 51.23 → 0.60 lower because of PV(dividends).

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Example 4 (Hightest / Unilever), pp. 90–91.

An equity index forward with continuous dividend yield uses $F_0(T) = S_0 \cdot e^{((r - i) T)}$ — the standard for index futures pricing

For indexes, dividends are treated as a CONTINUOUS yield i because the many constituents smear payment date. r − i is the net carry.

$$F_0(T) = S_0 \cdot e^{((r + c - i) T)} \quad \text{(for indexes } c = 0 \Rightarrow F_0 = S_0 \cdot e^{((r - i)T)}\text{)}$$

where: i = dividend yield (continuous); c = carry cost rate (continuous); r = risk-free rate (continuous).

VFO NIFTY 50 THREE-MONTH INDEX FORWARD

Spot NIFTY 50 = INR 15,200. Dividend yield i = 2.2%. INR risk-free r = 4.0%. T = 0.25y. Compute F_0.

1. Net carry rate = r − i = 4.0% − 2.2% = 1.8%
2. Exponent = 0.018 × 0.25 = 0.0045
3. $e^{0.0045} \approx 1.004510$
4. F_0 = 15,200 × 1.004510 = INR 15,268.55

F_0 = INR 15,268.55. If i > r, the forward would be BELOW spot — dividends outweigh opportunity cost.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Example 5 (VFO / NIFTY 50), pp. 91.

FX forwards price off covered interest-rate parity — the forward exchange rate is the spot rate grown at the interest-rate differential, not the level

Borrow in domestic, convert spot, lend in foreign, lock the FX at forward — two risk-free strategies must return the same, which forces F.

$$F_{0,\,f/d}(T) = S_{0,\,f/d} \cdot e^{(r_f - r_d)T}$$

where: f = price (foreign) currency (numerator); d = base (domestic) currency (denominator). Forward quote in same convention as spot.

AUD/USD 6-MONTH FORWARD — IRP IN ACTION

Spot S_0 = 1.3335 AUD/USD (f = AUD, d = USD). 6m AUD r_f = 0.05%; 6m USD r_d = 0.20%. T = 0.5y.

1. Interest-rate differential ($r_f - r_d$) = 0.0005 − 0.0020 = −0.0015 p.a.
2. Exponent = −0.0015 × 0.5 = −0.00075
3. $e^{-0.00075} \approx 0.999250$
4. F_0 = 1.3335 × 0.999250 = 1.3325 AUD/USD

F_0 = 1.3325. Fewer AUD are needed to buy USD in the future → AUD at a forward PREMIUM (the currency with the LOWER rate).

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Example 6 (AUD/USD replication), pp. 91–93.

Interest-rate parity means the currency with the LOWER risk-free rate trades at a forward PREMIUM — arbitrage would eat any deviation in seconds

If F deviates from spot × e^((rf − rd)T), a three-legged trade (spot + two money-market legs) locks in riskless profit.

$$F_{KRW/EUR}(T) = S_{KRW/EUR} \cdot e^{(r_{KRW} - r_{EUR})T}$$

MONTAU AG — KRW 650m EXPORT CONTRACT IN 75 DAYS

Spot S_0 = 1,300 KRW/EUR. r_KRW = 0.75%. r_EUR = −0.25%. T = 75/365 year.

1. Rate differential (r_KRW − r_EUR) = 0.0075 − (−0.0025) = 0.0100
2. Exponent: 0.0100 × (75/365) = 0.002055
3. F_0 = 1,300 × e^0.002055 = 1,300 × 1.002057 = 1,302.67 KRW/EUR
4. Verify via arbitrage: convert KRW 650m at spot → EUR 500,000; grow at −0.25% for 75 days → EUR 499,743.22
5. OR grow KRW 650m at 0.75% for 75 days → KRW 651,002,484.59
6. Implied forward = 651,002,484.59 / 499,743.22 = 1,302.67 ✓

F_0 = 1,302.67 KRW/EUR. KRW trades at forward DISCOUNT (higher rate). Arbitrage pins it.

Source: CFA Institute, Curriculum Vol 7, Learning Module 4, Example 7 (Montau KRW/EUR), pp. 93–94.

Futures are forwards with daily mark-to-market and margining — the cumulative P&L matches a forward's, but the cash-flow profile is radically different

Each day the futures MTM is settled to cash and the contract value RESETS to zero. Cumulative MTM through T ≈ final forward payoff.

$$\text{Daily MTM (long)} = [\, f_t(T) - f_{t-1}(T) \,] \times \text{contract size}$$

where: $f_t(T)$ = futures settlement price on day t; contract is re-struck to zero after settlement.

CFA LM 6 PP#2 — SHORT CRUDE OIL FUTURES, DAY T−5

Investor is short 1,000 barrels of crude at initial futures price f_0 = USD 69.00. On day T−6 settlement was 66.02; on day T−5 it rose to 68.01.

1. Daily futures price change: $f_{T-5} - f_{T-6}$ = 68.01 − 66.02 = +1.99/bbl
2. MTM to LONG = +1.99 × 1,000 = +USD 1,990
3. MTM to SHORT = −1.99 × 1,000 = −USD 1,990 (paid out via margin)
4. Short's margin account is debited USD 1,990 that day; may trigger a margin call below maintenance level.

Short MTM loss on day T−5 = USD 1,990 — settled that evening, not at maturity.

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, "Pricing of Futures Contracts", pp. 132–140.

At maturity the futures price converges to spot — basis = S − F grinds to zero and the hedger's payoff is anchored

Basis = S − F converges to 0 at T by no-arbitrage (for a cash-settled contract). Before T, basis moves with r and carry.

$$f_0(T) = [S_0 + PV_0(C)](1+r)^T \qquad Basis_t = S_t - f_t(T) \to 0 \text{ as } t \to T$$

CME COPPER FUTURES — MARGIN-CALL TRIGGER

Contract size 25,000 lb. S_0 = USD 4.25/lb. Storage cost USD 10/contract at T. r = 1.875%. T = 1/12. Initial margin USD 10,000; maintenance USD 6,000.

1. PV(C) per contract: 10 × (1.01875)^−(1/12) = USD 9.98
2. Compute futures price: f_0 = [(4.25 × 25,000) + 9.98] × (1.01875)^(1/12)
3. f_0 = (106,250 + 9.98) × 1.001549 = USD 106,425/contract ≈ USD 4.257/lb
4. Margin-call triggered when loss = 10,000 − 6,000 = USD 4,000
5. USD 4,000 / 25,000 lb = USD 0.16/lb → call at price 4.257 − 0.16 = USD 4.10/lb (≈ USD 102,425/contract)

Margin call if price drops below USD 4.10/lb — basis and storage built into the 4.257 no-arb price.

[1] Basis risk = residual risk that the derivative underlying and the hedged exposure don't move in lockstep before maturity.

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, Practice Problem 1 (CME copper), pp. 146–148.

Forwards and futures prices match under constant or uncorrelated rates; they diverge via the convexity bias when rates and prices are correlated

Practically tiny for short-dated contracts; non-trivial for multi-year interest-rate contracts (Eurodollar / SOFR futures vs IRS).

WHEN DO PRICES DIVERGE?

- EQUAL when r is constant OR r uncorrelated with f.
- Futures > forward when f POSITIVE-correlated with r — long futures re-invests MTM gains at the higher rate.
- Futures < forward when f NEGATIVE-correlated with r — e.g., bond futures (bond price falls when r rises).
- The more desirable contract tends to have the higher price.

CONVEXITY BIAS + CCP

- FRA payoff is DISCOUNTED at MRR (PV); futures payoff is LINEAR in the quoted rate.
- This convexity gap makes futures MORE attractive than FRAs for the long — futures prices above FRA-implied.
- Central-counterparty clearing for standardized OTC swaps imposes futures-like daily margin → shrinks the practical gap between OTC forward and futures pricing.
- Dealers pass margin requirements to end-users — financing and collateral costs matter.

Source: CFA Institute, Curriculum Vol 7, Learning Module 6, "Forward and Futures Price Differences", pp. 140–144.

SECTION 05

Swaps

A swap is a series of firm commitments — equivalent to a strip of forwards, but with a CONSTANT fixed rate instead of period-specific forward rates

One contract with ONE fixed rate replaces N FRAs with N different fixed rates — fewer tickets, same aggregate exposure.

$$\text{Periodic settlement (pay-fixed)} = (\text{MRR}_{t-1} - s_N) \times \text{Notional} \times \text{Period}$$

where: s_N = fixed swap rate for N-period swap; MRR sets at start of period; paid at end.

STRIP OF FRAs

Each FRA has its OWN fixed rate equal to the implied forward rate for that period. Administrative burden: N contracts, N rates to monitor, N counterparty exposures.

INTEREST-RATE SWAP

ONE contract with ONE constant fixed rate (s_N) that equates aggregate PV of fixed and floating legs at t = 0. Same economic exposure, far simpler to transact and manage.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Swaps vs Forwards", pp. 152–155.

The par swap rate is the single fixed rate that makes PV of expected floating cash flows equal PV of fixed cash flows at t=0 — a breakeven IRR on the forward curve

s_N is the internal rate of return that equates the forward-rate-implied floating stream to a level fixed stream over N periods.

$$\Sigma_{i=1..N} \; IFR_i / (1 + z_i)^i \;=\; s_N \times \Sigma_{i=1..N} \; 1 / (1 + z_i)^i$$

where: IFR_i = implied forward rate for period i (from the zero curve); z_i = zero rate for maturity i; s_N = par swap rate.

THE INTUITION IN TWO LINES

LEFT side = PV(floating) — each IFR_i discounted back at the matching zero rate. RIGHT side = PV(fixed) — the level swap rate s_N multiplied by the sum of discount factors. Setting left = right and solving for s_N gives the par swap rate.

Geometrically: s_N is a horizontal line drawn through the forward curve such that the PV-weighted area above equals the PV-weighted area below. Slope-up curves push s_N above the first IFR; slope-down curves pull it below.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Par Swap Rate", pp. 155–158.

Worked: a 3-year IRS prices at s_3 = 3.9641% given the zero curve 2.396% / 3.4197% / 4.0005%

Step 1 derive IFRs from zero curve. Step 2 plug into par-swap equation. Step 3 solve for s_N — the output equals the flat-coupon par bond yield.

Period	Zero rate z_i	Implied forward IFR_i	Derivation
1	2.3960%	2.3960%	IFR_{0,1} = z_1
2	3.4197%	4.4536%	$(1.034197)^2 / (1.02396) - 1$
3	4.0005%	5.1719%	$(1.040005)^3 / (1.034197)^2 - 1$

SOLVE FOR s_3

PV(float) = Σ IFR_i / (1+z_i)^i ; PV(fixed) = s_3 × Σ 1/(1+z_i)^i. Set equal, solve.

1. PV(float) = $2.3960/1.02396 + 4.4536/(1.034197)^2 + 5.1719/(1.040005)^3 = 0.111017$
2. Sum of discount factors Σ 1/(1+z_i)^i = 2.800545

s_3 = 3.9641% — note s_3 lies BETWEEN the 2.396% front IFR and the 5.172% back IFR, as expected on an upward curve.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, Example 1 (Swaps as Combination of Forwards), pp. 153–156.

At inception V_0(swap) = 0; on any settlement date the swap is worth the current settlement plus the PV of all remaining future exchanges

At t=0 PV(fixed) = PV(floating) by construction. As time and rates move, the two legs drift apart — that delta is the swap MTM.

V_t (swap) = Current period settlement + PV_t(remaining fixed vs floating exchanges)

where: Settlement = (MRR − s_N) × Notional × Period for pay-fixed; opposite sign for receive-fixed.

ESTERR INC. — CAD 250m PAY-FIXED IRS AT 2.05% Q

Esterr pays s = 2.05% quarterly, receives 3m CAD MRR. First MRR = 0.50%. Notional 250m. Period = 0.25.

1. Period-1 settlement = (0.50% − 2.05%) × 250m × 0.25 = −CAD 968,750 (Esterr pays)
2. Period-2 settlement (MRR = 0.74% expected) = (0.74% − 2.05%) × 250m × 0.25 = −CAD 818,750
3. Because the forward curve is upward sloping, future expected MRRs exceed 2.05% → future settlements swing positive for Esterr.
4. As time passes with unchanged forward curve, PV(floating received) > PV(fixed paid) → Esterr MTM gain.

V_0 = 0 at inception; period-1 cash outflow CAD 968,750; but aggregate swap MTM turns positive as upward curve delivers rising MRRs.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, Example 3 (Esterr Swap Value), pp. 159–162.

An interest-rate swap is economically equivalent to a pair of bonds — pay-fixed ≡ long FRN + short fixed-rate bond; receive-fixed ≡ long fixed + short FRN

Principals cancel at inception and at maturity. Only the coupon legs remain — so a swap is simply two bond coupons swapped with a partner.

Swap side	Equivalent bond portfolio	Effective duration	Higher r
Pay-fixed / Receive-floating	LONG FRN (priced at par at each reset) + SHORT fixed bond	Negative (~ short bond)	MTM GA
Receive-fixed / Pay-floating	LONG fixed bond + SHORT FRN	Positive (~ long bond)	MTM LO

WHY THE PRINCIPALS CANCEL

At t = 0 a portfolio manager borrows notional N at MRR (short FRN = +N cash) AND lends N at the fixed swap rate (long fixed bond = −N cash). Net cash flow at t = 0 = 0. Ditto at maturity: repay N at MRR (−N) and receive N at fixed (+N), net = 0. What remains between these dates is just the coupon leg exchange — identical to a pay-fixed swap. This is the bedrock intuition used across fixed-income trading: view an IRS as the rate differential between two bonds, and the duration / DV01 of the swap is the DIFFERENCE in duration / DV01 of those two bonds.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Using Swaps to Manage Fixed Income", pp. 157–158.

Post-inception, a rise in expected forward rates raises PV(floating), producing an MTM gain for the fixed-rate payer and a loss for the fixed-rate receiver

Rule: expected rates UP ⇒ pay-fixed WINS. Expected rates DOWN ⇒ receive-fixed WINS. Same sign as a position in a fixed-rate bond.

Event post-inception	Effect on PV(float)	Effect on PV(fixed)	Pay-fixed MTM	Receive-fixed MTM
Forward curve shifts UP parallel	Up	Down (higher disc.)	GAIN	LOSS
Forward curve shifts DOWN parallel	Down	Up (lower disc.)	LOSS	GAIN
Time passes, upward curve unchanged	PV(float) > PV(fix)	—	GAIN	LOSS
Time passes, downward curve unchanged	PV(float) < PV(fix)	—	LOSS	GAIN

DURATION INTUITION

A receive-fixed swap behaves like a long fixed-rate bond — positive duration. Active managers who want to INCREASE portfolio duration without buying more bonds go receive-fixed; managers who want to CUT duration go pay-fixed. The IRS has become the primary duration-management tool in institutional fixed-income books.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, "Swap Values and Prices", pp. 159–165.

Worked: Fyleton receives fixed 2.38% on a 5y GBP 200m IRS and the forward curve falls — MTM gain equivalent to a long fixed-rate bond rallying

Receive-fixed = long-bond proxy. Rates fall → bond price rises → receive-fixed swap MTM positive. Same directional exposure, far cheaper to execute.

Period settlement (receive-fixed) = (s_N − MRR) × Notional × Period

FYLETON INVESTMENTS — RECEIVE-FIXED 5y GBP 200m

Fyleton receives fixed s = 2.38% semi-annually, pays 6m GBP MRR. Initial MRR sets at 0.71%. Period = 0.5. Upward-sloping initial forward curve.

1. First settlement at t = 0.5: (2.38% − 0.71%) × 200m × 0.5
2. = 0.0167 × 200,000,000 × 0.5 = GBP 1,670,000 RECEIVED
3. Then forward curve shifts DOWN — PV(floating paid) falls faster than PV(fixed received).
4. Net: PV(fixed received) > PV(floating paid) → MTM GAIN for Fyleton.
5. Risk profile mirrors a LONG fixed-rate bond: rates fall → bond rallies → swap MTM gains.

First exchange: +GBP 1,670,000 to Fyleton. Subsequent rate cuts compound the MTM gain — classic long-duration profile.

Source: CFA Institute, Curriculum Vol 7, Learning Module 7, Example 4 (Fyleton Investments), pp. 162–164.

KEY TAKEAWAYS

Deck 17 — Derivatives I: Forwards, Futures & Swaps

- Derivatives derive value from an underlying; firm commitments (forwards, futures, swaps) have symmetric payoffs and $V_0 = 0$, while contingent claims (options, CDS) have asymmetric payoffs and a premium.
- No-arbitrage engine: $F_0(T) = S_0(1+r)^T$ for assets without carry; extend to $F_0 = [S_0 - PV(I) + PV(C)](1+r)^T$ for dividends, coupons, storage costs.
- FX covered interest-rate parity: $F_{f/d} = S_{f/d} \cdot e^{(rf - rd)T}$. The currency with the LOWER risk-free rate trades at a forward PREMIUM. Montau KRW/EUR: 1,300 → 1,302.67.
- Par swap rate s_N solves $\Sigma\, IFR_i / (1+z_i)^i = s_N \times \Sigma\, 1/(1+z_i)^i$. 3-year example: $s_3 = 3.9641\%$ on zero curve (2.396 / 3.4197 / 4.0005)%.
- OTC = customized bilateral under ISDA, with counterparty risk; ETD = standardized, centrally cleared, margined daily. Post-crisis CCP clearing has narrowed the gap on standardized OTC.
- Forward MTM over life: $V_t = S_t - F_0(T)(1+r)^{-(T-t)}$ for long — sensitive to both S_t and r. Biomian: INR 300.84 forward → 12.90 MTM at t = 0.25 when S_t = 285.
- Futures = forwards with daily MTM + margining. Basis S − F converges to 0 at T. Convexity bias makes futures > forwards when f is positive-correlated with r; central clearing of OTC narrows the practical gap.
- IRS ≡ pair of bonds: pay-fixed ≡ long FRN + short fixed bond (negative duration); receive-fixed ≡ long fixed + short FRN (positive duration). Fyleton receive-fixed on falling rates = bond rally.